

# Musculoskeletal & Integumentary systems And Craniofacial Region

## Test 1 (50 Qs) - ( Lower limb 18 Qs, Histology 14Qs, Embryology 18 Qs )

### Question 1

The tendon of which muscle passes through the tarsal tunnel and closest posteriorly to medial malleolus?

Select one:

- a. Flexor hallucis longus
- b. Peroneal tertius
- c. Flexor digitorum longus
- d. Tibialis posterior
- e. Tibialis anterior

**ANSWER: D**

### Question 2

Apocrine sweat glands are:

Select one:

- a. Compound tubuloacinar
- b. Simple branched alveolar

- c.Simple branched alveolar
- d.Simple coiled tubular
- e.Compound tubular

**ANSWER: D**

#### Question 3

Which bony marking is used to distinguish the femoral hernia from the inguinal hernia?

Select one:

- a.pubic crest
- b.greater trochanter of femur
- c.pubic tubercle
- d.lessor trochanter of femur
- e.ischial spine

**ANSWER: C**

#### Question 4

Housemaid's knee is an inflammation of which bursa?

Select one:

- a.semimembranosus
- b.suprapatellar
- c.anserinus
- d.prepatellar
- e.infrapatellar

**ANSWER: D**

Question 5

Injury to medial collateral ligament is likely to cause damage to which other component of the knee joint?

Select one:

- a.posterior cruciate ligament and medial meniscus
- b.anterior and posterior cruciate ligament
- c.medial meniscus and anterior cruciate ligament
- d.anterior cruciate ligament and lateral meniscus
- e.posterior cruciate ligament and lateral meniscus

**ANSWER: C**

Question 6

Which of the following statements is correct about the cells of cartilage?

Selectone:

- a.Chondroblasts have processes which communicate by gap junction with eachother.
- b.Chondrocytes located peripherally are round, while those located deeper are oval.
- c.Chondroblasts are found in the lacunae.
- d.Chondrogenic cells can differentiate into chondroblasts as well as osteogenic cells
- e.Chondrocytes have eosinophilic cytoplasm.

**ANSWER: D**

Question 7

Which of the following limb defects are caused by constrictive action of amniotic bands?

Selectone:

- a.Fused digits

- b.Amputation
- c.Polydactyl
- d.Cleftfoot
- e.Phecomalia

**ANSWER: B**

#### Question 8

After a lesion of the tibial part of sciatic nerve, some active flexion may still be possible at the knee joint; the muscles responsible for this remaining flexion is:

Select one:

- a.Plantaris muscle
- b.short head of Biceps femoris
- c.Gastrocnemius muscle
- d.Popliteus muscle
- e.long head of Biceps femoris muscle

**ANSWER: B**

#### Question 9

Which statement is TRUE regarding the knee joint?

Select one:

- a.the anterior cruciate ligament is slacked when the knee joint extended and taut when it was flexed
- b.both the lateral and medial menisci have thick inner border
- c.the lateral meniscus is strongly attached to the lateral collateral ligament
- d.the posterior cruciate ligament that prevents posterior displacement of tibia

e.C and D

**ANSWER: D**

Question 10

Which type of the following cells are located in the Hawship's lacuna?

Select one:

- a.Osteocyte
- b.Chondroblast
- c.Chondrocyte
- d.Osteoblasts
- e.Osteoclast

**ANSWER: E**

Which of the following is secreted from the lateral plate mesoderm to initiate limb growth?

Select one:

- a.Fibroblast Growth Factor-4(FGF-4)
- b.Sonic hedgehog(SHH)
- c.Fibroblast Growth Factor-10(FGF-10)
- d.Radical fringe
- e.Bone Morphogenesis Proteins

**ANSWER: C**

Question 12

All of the following embryonic tissues are origins of the skeletal system, EXCEPT:

Select one:

- a.Lateral plate mesoderm
- b.Somitomeres
- c.Neural crest
- d.Paraxial mesoderm
- e.Pharyngeal pouch

**ANSWER: E**

#### Question 13

The common fibular nerve can be palpated in which region of the knee?

Select one:

- a.as it crosses the lateral condyle of tibia
- b.as it winds around the medial side of the neck of fibula
- c.as it winds around the lateral side of the neck of fibula
- d.as it passes around the medial condyle of tibia
- e.as it passes forward between the tibia and fibula

**ANSWER: C**

#### Question 14

The sarcoplasmic reticulum of skeletal muscle cells are made by:

Select one:

- a.Pinosomes.
- b.Rough endoplasmic reticulum.
- c.Golgi apparatus
- d.Smooth endoplasmic reticulum.
- e.Polyribosomes

**ANSWER: D**

Question 15

Which of the following birth defects shows short neck, low hairline and restriction of neck movements?

Select one:

- a. Hemi-vertebra
- b. Spina bifida occulta
- c. Klippel-Feil Syndrome
- d. Scoliosis
- e. Spina-bifida cystica

**ANSWER: C**

Question 16

Which of the following birth defects is a result of Cranioschisis?

Select one:

- a. Anencephaly
- b. Achondroplasia
- c. Acrocephaly
- d. Plagiocephaly
- e. Scaphlocephaly

**ANSWER: A**

Question 17

Which of the following muscles is derived from paraxial mesoderm?

Select one:

- a.Cardiac muscle
- b.Pupillary muscle
- c.Skeletal muscle
- d.Smooth muscle
- e.Mammary gland muscle

**ANSWER: C**

Question 18

Which of the following nerves is formed by a combination of the dorsal segmental branches of spinal nerves?

Select one:

- a.Ulnar
- b.Musculocutaneous
- c.Radial
- d.Median
- e.Tibial

**ANSWER: C**

Question 19

Which of the following somitomeres is an origin for jaw closing muscles?

Select one:

- a.1
- b.4
- c.6
- d.7

e.3

**ANSWER: B**

Question 20

Incorrect statement regarding the epimeric musculature,

Select one:

- a. Supplied by dorsal primary rami nerves
- b. Forms the back musculature
- c. Expresses muscle specific gene MYO-D
- d. Involves muscle specific gene MYF5
- e. Migration of dorsomedial cells of somites

**ANSWER: C**

Question 21

All of the following structures are passed through the greater sciatic foramen below piriformis muscle EXCEPT:

Select one:

- a. superior gluteal nerve
- b. inferior gluteal vessels
- c. posterior cutaneous nerve of thigh
- d. sciatic nerve
- e. inferior gluteal nerve

**ANSWER: A**

Question 22

Which of the following bones is formed by membranous ossification?

Select one:

- a.Chordal neurocranium
- b.Viscerocranium
- c.Cranial vault
- d.Prechordal neurocranium
- e.Cranial base

**ANSWER: C**

Question 23

Lymph from the skin of big toe and medial side of foot drains directly into:

Select one:

- a.external inguinal lymphnode
- b.vertical group of superficial inguinal lymphnodes
- c.horizontal group of superficial inguinal lymphnodes
- d.popliteal lymphnodes
- e.internal iliac lymphnodes

**ANSWER: B**

Question 24

The chief arterial blood supply for femoral head in children is:

Select one:

- a.lateral circumflex femoral artery
- b.artery to head of femur
- c.perforating arteries

d.popliteal artery

e.medial circumflex femoral artery

**ANSWER: B**

Question 25

Anterior thigh muscles are mainly\_\_\_at kneejoint

Select one:

a.rotator

b.adductor

c.extensor

d.flexor

e.abductor

**ANSWER: C**

Question 26

Which of the following statements is not correct regarding to the bone?

Select one:

a.Spongy bone contains marrow spaces.

b.Compact bone has outer and inner circumferential laminae known as the Haversian system.

c.Cellbodies of the osteocytes are found within lacunae.

d.Canaliculi do not cross cement lines.

e.Resting osteoblasts form bone lining cells.

**ANSWER: B**

Question 27

All of the following syndromes can result mid facial defects, except:

Select one:

- a. Pfeiffer
- b. Crouzon
- c. Apert
- d. Cranioschisis
- e. Achondroplasia

**ANSWER: E**

Question 28

The sciatic nerve arises from which of the following segment of spinal cord?

Select one:

- a. L5, S1, S2 and S3
- b. L4, L5, S1 and S2
- c. L2 and L3
- d. L2, L3 and L4
- e. L4, L5, S1, S2 and S3

**ANSWER: E**

Question 29

Concerning the sarcomere of skeletal muscle fibers, which statement is true?

Select one:

- a. The Z lines bisect the A bands.
- b. The M lines bisect the H bands.
- c. The I bands represent the dark lines as seen under light microscope.

d.The diameters of A bands decrease when muscles contract.

e.It is the region between successive M lines.

**ANSWER: B**

Question 30

A newly deposited uncalcified bone matrix is known as:

Select one:

a.Perosteoid

b.Cancellousbone

c.Osteon

d.Haversian system

e.Osteoid

**ANSWER: E**

Outer coat of hair follicle is:

Select one:

a.Absent from fine hairs.

b.Equivalent to the most inner layer of epidermis.

c.Made by dead cells known as cuticle.

d.Made by connective tissue continuous with the dermis.

e.Equivalent to stratum corneum.

**ANSWER: D**

Question 32

All of the following are found in the second layer of foot EXCEPT?

Select one:

- a.tendon of flexor hallucis longus
- b.tendon of flexor digitorum longus
- c.adductor hallucis
- d.quadratus plantae
- e.lumbricals

**ANSWER: C**

Question 33

Which of the following bones arises from the ventral portion of the first pharyngeal arch?

Select one:

- a.Maxilla
- b.Temporal
- c.Mandible
- d.Temporal
- e.Zygomatic

**ANSWER: C**

Question 34

Which statement is FALSE regarding femoral triangle?

Select one:

- a.femoral nerve is a content within a femoral sheath
- b.the lateral border of femoral triangle is formed by sartorius muscle
- c.the contents of femoral triangle from lateral to medial are femoral nerve, artery, vein and canal

- d.femoral ring is the proximal opening in the femoral canal
- e.the superior border of femoral triangle is formed by inguinal ligament

**ANSWER: A**

Question 35

Which of the following disorders is caused by a loss of melanocytes?

Select one:

- a.Piebaldism
- b.Albinism
- c.Occulocutaneous albinism
- d.Vitiligo
- e.Waardenburg syndrome

**ANSWER: D**

Which of the following cell attachments are normally found along the longitudinal segments of the intercalated discs of the cardiac muscle?

Select one:

- a.Gap junctions
- b.Zonula adherens
- c.Hemidesmosomes
- d.Macula adherens
- e.Zonula occludens

**ANSWER: A**

Question 37

True statement regarding the adductor canal and hiatus:

Select one:

- a.the posterior border of adductor canal is formed by adductor longus and magnus
- b.the medial border of adductor canalis formed by vastus medialis muscle
- c.saphenous nerve passes through the adductor hiatus
- d.the adductor hiatus is a space within adductor longus muscle
- e.the lateral border of adductor canalis formed by sartorius muscle

**ANSWER: A**

Question 38

All the following arteries are involved in cruciate anastomosis EXCEPT:

Select one:

- a.Superior gluteal artery
- b.Lateral circumflex femoral artery
- c.Inferior gluteal artery
- d.First perforating artery
- e.Medial circumflex femoral artery

**ANSWER: A**

Question 39

Which of the following forms nucleus pulposus of the intervertebral disc?

Select one:

- a.Costal processes

b.Mesenchymal centrum

c.Neural arch

d.Cranial sclerotome

e.Notochord

**ANSWER: E**

#### Question 40

With regard to cartilage, which statement is false?

Select one:

a.Fibrocartilage is present in regions subjected to high tension.

b.Hyaline cartilage is characterized by anamorphous glassy matrix.

c.Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.

d.Fibro cartilage has thick bundles of collagen fibres in the matrix.

e.Fibrocartilage is used for endochondral bone ossification

**ANSWER: E**

#### Question 41

Melanocytes are located in which layer of the epidermis?

Select one:

a.Stratum spinosum

b.Stratum basale

c.Stratum lucidum

d.Stratum corneum

e.Stratum granulosum

**ANSWER: B**

Question 42

Endochondral bone ossification includes the following, except:

Select one:

- a. Chondrocytes changing to osteocytes.
- b. Proliferation of cartilage cells.
- c. Calcification of the cartilage matrix.
- d. Interruption of diffusion to the cartilage at the area of primary ossification.
- e. A cartilage model.

**ANSWER: A**

Question 43

Which femoral bony landmark represents the anterior attachment of the hip joint capsule?

Select one:

- a. intertrochanteric crest
- b. linea aspera
- c. intertrochanteric line
- d. pectineal line
- e. gluteal tuberosity

**ANSWER: C**

Question 44

Region of the sarcomer containing only the thin myofilaments is represented by:

Select one:

- a. Z disc

b.A bands

c.M lines

d.H bands

e.I bands

**ANSWER: E**

#### Question 45

Which of the following structures induces proliferation of cells in the progress zone of the developing limb?

Select one:

a.Paraxial mesoderm

b.Apical ectodermal ridge

c.Neural crest cells

d.Intermediate mesoderm

e.Lateral plate mesoderm

**ANSWER: B**

#### Question 46

Which of the following lies on the ventromedial position of the somites?

Select one:

a.Sclerotome

b.Intermediate mesoderm

c.Dermomytome

d.Myotome

e.Endoderm

**ANSWER: A**

Question 47

Which type of cells act as antigen presenting cells in the epidermis?

Select one:

- a.Mast cell
- b.Langerhan's cells
- c.Keratinocytes
- d.Merkel's cells
- e.Melanocytes

**ANSWER: B**

Question 48

Suppose you are sitting on the chair for your mid-exam, which bony landmark is support you while you are sitting?

Select one:

- a.pubic tubercle
- b.ischial tuberosity
- c.ischial spine
- d.ilic tuberosity
- e.ilic crest

**ANSWER: B**

Question 49

Which of the following fontanelles of the skull of a newborn closes at about 3months of age

through membranous ossification?

Select one:

- a. Anterolateral
- b. Mastoid
- c. Posterior
- d. Anterior
- e. Posterolateral

**ANSWER: C**

Question 50

Which of the following muscles is formed from the hypoblasts of cervical hypomeres?

Select one:

- a. Erector spinae
- b. Scalene
- c. External intercostals
- d. Rectus abdominus
- e. Trapezius

**ANSWER: B**

**TEST 2 ( 50Qs) (Head & Neck anatomy -  
Neck anatomy part, covered by sorressa,  
20 Questions repeated from the  
previous/2010/ batch)**

---

**TEST 3**

---

**TEST 4**

---

# Final exam

## Question 1

Possible adverse effect from high dose oxymetazoline

Select one:

- a. Increased intraocular pressure
- b. Hypotension
- c. Hypertension
- d. Sedation
- e. Miosis

**ANSWER: B**

## Question 2

The latch-bridge mechanism in smooth muscle is responsible for

Select one:

- a. Sustained muscle contraction
- b. Unstable membrane potential
- c. Excitation-contraction coupling
- d. Fast muscle twitch

e. Rhythmic contraction

**ANSWER: A**

### Question 3

All the following structures are related with labyrinthine(medial)

wall of tympanic cavity except:

Select one:

- a. Tympanic plexus
- b. Chorda tympani nerve
- c. Tympanic nerve
- d. Lesser petrosal nerve
- e. Facial nerve

**ANSWER: B**

### Question 4

All of the following statements are correct, EXCEPT

Select one:

- a. Myasthenia gravis is caused by an autoimmune destruction of voltage gated  $\text{Ca}^{2+}$  channels on the presynaptic motor nerve.
- b. Damage to lower motor neurons results in flaccid paralysis
- c. The strength of cardiac muscle contraction depends to a great extent on the concentration of  $\text{Ca}^{2+}$  in the ECF
- d. Damage to upper motor neurons results in flaccid paralysis followed by spastic paralysis in a few weeks

e. Skeletal muscle contraction is caused almost entirely by  $\text{Ca}^{2+}$  released from the sarcoplasmic reticulum inside the skeletal muscle fiber itself

**ANSWER: A**

#### Question 5

To test the normal function of inferior oblique muscle of one eye, what would you have a patient do?

Select one:

- a. Depress and abduct the eye
- b. Elevate and medially rotate the eye
- c. Elevate and laterally rotate the eye
- d. Elevate and adduct the eye
- e. Depress and adduct the eye

**ANSWER: C**

#### Question 6

Which of the following tooth abnormalities is caused by consumption of tetracyclines by pregnant mothers and children?

Select one:

- a. Eruption at birth
- b. Discoloration
- c. Abnormal shape and size
- d. Deficient in enamel
- e. Abnormal in number

**ANSWER: B**

Question 7

Which of the following statements is NOT CORRECT regarding glycogen metabolism during exercise?

Select one:

- a. During exercise muscle glycogen is converted into glucose than glucose-6-P
- b. Utilization of glycogen is lower in trained athletes
- c. Glycogen depletion is associated with fatigue
- d. Glycogen is the primary source of fuel during exercise lasting 60-90 minutes
- e. Rate of glycogen utilization is higher during early exercise

**ANSWER: A**

Question 8

Which of the following passes through the gap between the superior and middle constrictors?

Select one:

- a. levatorveli palatini
- b. glossopharyngeal nerve
- c. internal laryngeal nerve
- d. superior laryngeal artery
- e. superior laryngeal vein

**ANSWER: B**

Question 9

Which one is not correct concerning Enterococci:

Select one:

- a. Enterococci are non-haemolytic Streptococci
- b. Enterococci are normal flora of the intestinal tract
- c. They cause Urinary tract infections (UTI)
- d. They cause alpha-haemolysis on blood agar
- e. faecalis and E. facium are the 2 major pathogens of humans

**ANSWER: D**

Question 10

The nerve of the pterygoid canal does NOT innervate which of the following?

Select one:

- a. Lacrimal gland
- b. Mucosal glands of nasal cavity
- c. Mucosal glands of upper pharynx
- d. Submandibular glands
- e. Palatine glands

**ANSWER: D**

Question 11

Re-infestation with *Sarcoptes scabiei* produces symptoms quicker than primary infestation. This is because:

Select one:

- a. It is a function of immune memory
- b. The skin loses its strength and innate resistance after primary infestation,

- c. There is often a lot more organic matter in the skin burrows that can support habitation of *Sarcoptes scabiei*,
- d. There are some surviving ova and adult stages in skin burrows that facilitate infestation,
- e. The infective dose is higher during re-infestation

**ANSWER: A**

#### Question 12

Regarding gas gangrene one of the following is correct:

Select one:

- a. The clinical features are due to the release of protein endotoxin
- b. Gas is invariably present in the muscle compartments
- c. Infection is by ingestion of contaminated food
- d. Clostridial species are Gram-negative spore forming anaerobes
- e. It is caused by *Clostridium Botulinum*

**ANSWER: B**

#### Question 13

Which of the following limb defects can be caused by teratogenic effect of thalidomide?

Select one:

- a. Amputation
- b. Brachydactyl
- c. Amelia
- d. Polydactyl

e. All of the above

**ANSWER: C**

#### **Question 14**

Wrong statement about the skeletal muscle

Select one:

- a. Skeletal muscles that rely mainly on oxidative phosphorylation have high capillary density
- b. Skeletal muscle fibers that rely mainly on glycolysis are fatigue resistant
- c. Can be tetanized
- d. Increased neuromuscular coordination produces greater force of contraction
- e. Acetylcholine is the only neurotransmitter which can excite the skeletal muscle

**ANSWER: B**

#### **Question 15**

The correct match between paranasal sinuses and their site of opening on the lateral wall of the nasal cavity is

Select one:

- a. Nasolacrimal duct — middle meatus
- b. Maxillary sinus — posterior end of hiatus semilunaris
- c. Middle ethmoidal air cells — bulla ethmoidalis
- d. Sphenoidal sinus — sphenoethmoidal recess
- e. Frontal sinus — upper end of hiatus semilunaris

Question 16

Which of the following muscles is correctly matched with its action?

Select one:

- a. Musculus uvulae . . . pulls uvula inferiorly
- b. Palatopharyngeus ... pulls walls of pharynx inferiorly, posteriorly, and laterally during swallowing
- c. Tensor veli palatini . . . opens auditory tube during swallowing and yawning
- d. Palatoglossus . . . depresses posterior part of tongue and draws soft palate away from tongue
- e. Levator veli palatini . . . depresses soft palate during swallowing and yawning

**ANSWER: C**

Question 17

Tinea nigra is caused by:

Select one:

- a. candida albicans
- b. Sporothrixschenkii
- c. Malssezia furfur
- d. Piedra hortae
- e. Exophila wernekii

**ANSWER: E**

Question 18

The most common localized benign neoplasms of synovium/tendon sheath is

Select one:

- a. Giant cell tumor of the tendon sheath
- b. Epithelioid sarcoma
- c. Synovial sarcoma
- d. Pigmented villonodular synovitis

**ANSWER: A**

Question 19

Injury to superficial peroneal nerve causes:

Select one:

- a. foot drop
- b. a shuffling gait
- c. loss of eversion of foot
- d. loss of inversion of foot
- e. a high skipping gait

**ANSWER: C**

Question 20

Which of the following prominences of the embryo forms the philtrum of upper lip, crest, and tip of nose?

Select one:

- a. Mandibular
- b. Medial nasal

- c. Lateral nasal
- d. Maxillary
- e. FrontonasalMaxillary

**ANSWER: B**

#### Question 21

The coagulase test is used to differentiate:

Select one:

- a. Streptococcus pyogenes from Enterococcus faecalis
- b. Staphylococcus aureus from Staphylococcus epidermidis
- c. Streptococcus pyogenes from Strept agalactiae
- d. Streptococcus pyogenes from Staphylococcus aureus
- e. Staphylococcus epidermidis from Neisseria meningitides

**ANSWER: B**

#### Question 22

25 year old patient coming to an emergency room had severe injury at middle of the SCM. Up on physical examination loss of cutaneous sensations from of anterior aspect of the neck. Which one of the following nerve is most commonly injured?

Select one:

- a. Transverse cervical nerve
- b. Great occipital nerve
- c. Great auricular nerve

- d. Supra - clavicular nerve
- e. Lesser occipital nerve

**ANSWER: A**

Question 23

Which one of the following is incorrect regarding carotid sinus?

Select one:

- a. Reacts to changes in arterial blood pressure
- b. Is a slight dilation at the origin of the internal carotid artery
- c. Innervated principally by the glossopharyngeal nerve
- d. The vagus nerve also supplies it
- e. Is a chemoreceptor

**ANSWER: E**

Question 24

Dermatophytosis:

Select one:

- a. restricted to developing countries only
- b. has annular appearance in skin infections
- c. not transmitted by contact
- d. re-infection is not possible
- e. Causes only skin infections

**ANSWER: B**

Question 25

The ability to look upward requires the simultaneous contractions of which two extraocular muscles?

Select one:

- a. Superior rectus and superior oblique
- b. Inferior rectus and superior oblique
- c. Superior oblique and inferior oblique
- d. Inferior rectus and inferior oblique
- e. Superior rectus and inferior oblique

**ANSWER: E**

Question 26

Bundles of spindle shaped cells arranged at right angles to one another "herring-bone pattern "is characteristic histologic pattern of:

Select one:

- a. Rhabdomyosarcoma
- b. Fibrosarcoma
- c. MFH
- d. Liposarcoma
- e. MPNST

**ANSWER: B**

Question 27

The large weight bearing joints of vertebral column, hips, and knee joints are most affected along with the distal interphalangeal joints

of the fingers in

Select one:

- a. Rheumatoid arthritis
- b. Osteoarthritis
- c. Tuberculous arthritis
- d. Gouty arthritis

**ANSWER: C**

Question 28

Selective alpha 2 agonist used to control intraocular pressure with minimal CNS effect

Select one:

- a. Clonidine
- b. Guanfacine
- c. Brimonidine
- d. Apraclonidine
- e. Guanabenz

**ANSWER: D**

Question 29

$\alpha$ 1A selective antagonist used for treatment of BPH (benign prostatic hyperplasia) with little effect on blood pressure

Select one:

- a. Alfuzosin
- b. Prazosin

- c. Terazosin
- d. Doxazosin
- e. Tamsulosine

**ANSWER: E**

#### Question 30

The latch-bridge mechanism in smooth muscle is responsible for

Select one:

- a. Unstable membrane potential
- b. Rhythmic contraction
- c. Excitation-contraction coupling
- d. Fast muscle twitch
- e. Sustained muscle contraction

**ANSWER: E**

#### Question 31

Which one of the following structures is NOT found in the adductor canal?

Select one:

- a. femoral artery
- b. femoral nerve
- c. saphenous nerve
- d. femoral vein
- e. nerve to vastus medialis

**ANSWER: B**

Question 32

Which of the following is present in the earliest phases of Paget's disease?

Select one:

- a. Bone deformities
- b. Densely sclerotic bone
- c. Foci of sclerosis with foci of osteolysis
- d. Predominance of osteolysis
- e. None of the above

**ANSWER: D**

Question 33

Which one of the following muscle has common nerve supply with posterior belly of digastric muscle?

Select one:

- a. Mylohyoid muscle
- b. Hyoglossus muscle
- c. Stylohyoid muscle
- d. Geniohyoid muscle
- e. Anterior belly of the digastric muscle

**ANSWER: C**

Question 34

All of the following except one are possible methods to prevent transmission or to control pediculosis resulting from body louse (Pediculus humanus corporis)

Select one:

- a. Health education about preventive methods
- b. Hygiene particularly at the individual level,
- c. Soaking of clothes in hot water,
- d. No nit policy in schools
- e. Prophylactic treatments

**ANSWER: E**

Question 35

Which of the following is true regarding tuberculous osteomyelitis?

Select one:

- a. It may rarely affect young adults
- b. Majority of cases are associated with active pulmonary tuberculosis
- c. The thoracic/ lumbar spines are most commonly affected
- d. Usually present with solitary bone involvement
- e. None of the above

**ANSWER: C**

Question 36

One of the following statements about transmission of onchocerciasis is not true:

Select one:

- a. Patients with generalized onchodermatitis are effectively infective to black flies,
- b. Localized lesions are good sources of mf for the feeding black flies,

- c. microfilarial (Mf) load affects intensity of transmission,
- d. Adult worms are not transmissible
- e. It is anthroponotic

**ANSWER: E**

Question 37

This prostaglandin analog is used to maintain the ductus arteriosus patent in some neonates awaiting cardiac surgery

Select one:

- a. Alprostadil
- b. Dinoprostone
- c. Mifepristone
- d. Misoprostol
- e. Latanoprost

**ANSWER: A**

Question 38

Which of the following disorders is caused by a loss of melanocytes?

Select one:

- a. Albinism
- b. Piebaldism
- c. Occulocutaneous albinism
- d. Vitiligo
- e. Waardenburg syndrome

**ANSWER: D**

Question 39

One of the following is not the effect of histamine

Select one:

- a. Powerful stimulant of sensory nerve endings (pain and itching)
- b. Causes bronchial smooth muscle dilation
- c. Decrease in systolic and diastolic blood pressure
- d. Causes contraction of intestinal smooth muscle
- e. Stimulant of gastric acid secretion

**ANSWER: B**

Question 40

Factors contributing to the stability of hip joint include all of the following EXCEPT:

Select one:

- a. the joint is the most stable when it is flexed and adducted
- b. the pectineus, adductors, gracilis and piriformis
- c. the gluteus medius, minimus and tensor fasciae latae
- d. the acetabular labrum
- e. the ligaments arising from the constituent bones of pelvis especially the iliofemoral ligament

**ANSWER: A**

Question 41

Which type of cells act as antigen presenting cells in the epidermis?

Select one:

- a. Merkel's cells
- b. Langerhan's cells
- c. Melanocytes
- d. Keratinocytes
- e. Mast cell

**ANSWER: B**

Question 42

Clonidine may increase blood pressure in

Select one:

- a. Patients with severe hypertension
- b. Patients with pure autonomic failure
- c. Patients taking non selective MAOIs
- d. Patients consuming tyramine containing foods
- e. Patients with moderate hypertension

**ANSWER: B**

Question 43

Which of the following lesions is caused by human papilloma virus (HPV) and has the worst outcome?

Select one:

- a. Verruca vulgaris
- b. Verruca plana

- c. Squamous cell carcinoma
- d. Condyloma acuminatum
- e. Verruca plantaris

**ANSWER: C**

**Question 44**

True statement regarding femoral canal:

Select one:

- a. it transmits the femoral branch of the genitofemoral nerve
- b. it is the lateral compartment of the femoral sheath
- c. it contains the femoral nerve
- d. it lies medial to the pubic tubercle
- e. it is medial to femoral vein

**ANSWER: E**

**Question 45**

Which of the following is the most common benign soft tissue tumor of adulthood?

Select one:

- a. Fibromatosis
- b. Neurofibroma
- c. Lipoma
- d. Nodular Fascitis
- e. Rhabdomyoma

**ANSWER: C**

Question 46

In the management of pheochromocytoma why are beta blockers used after the administration of alpha-blockers

Select one:

- a. To avoid competition at receptor site
- b. To avoid drug interaction at absorption site
- c. To avoid uncontrolled vasoconstriction
- d. To reduce tolerance for beta blockers
- e. To avoid excessive bradycardia

**ANSWER: C**

Question 47

Chronic infection involving cutaneous, subcutaneous and lymphatic tissue:

Select one:

- a. Sporotrichosis
- b. Mycetoma
- c. Chromoblastomycosis
- d. Lobomycosis
- e. Rhinosporidiosis

**ANSWER: A**

Question 48

Branches of the popliteal artery include all of the following EXCEPT:

Select one:

- a. inferior medial genicular artery

- b. superior lateral genicular artery
- c. middle genicular artery
- d. superior medial genicular artery
- e. descending genicular artery

**ANSWER: E**

#### Question 49

In adult the chief arterial supply to the head of femur is derived from which artery?

Select one:

- a. medial and lateral circumflex femoral artery
- b. obturator artery
- c. deep circumflex iliac artery
- d. internal pudendal artery
- e. superficial circumflex iliac artery

**ANSWER: A**

#### Question 50

All of the following statements are correct, EXCEPT

Select one:

- a. The strength of cardiac muscle contraction depends to a great extent on the concentration of  $\text{Ca}^{2+}$  in the ECF
- b. Skeletal muscle contraction is caused almost entirely by  $\text{Ca}^{2+}$  released from the sarcoplasmic reticulum inside the skeletal muscle fiber itself

- c. Damage to upper motor neurons results in flaccid paralysis followed by spastic paralysis in a few weeks
- d. Myasthenia gravis is caused by an autoimmune destruction of voltage gated  $\text{Ca}^{2+}$  channels on the presynaptic motor nerve.
- e. Damage to lower motor neurons results in flaccid paralysis

**ANSWER: D**

Question 51

Injury to ansa cervicalis during surgical reflection of the carotid sheath would result in paralyze of all of the following muscles of the neck except:

Select one:

- a. Thyrohyoid muscle
- b. Inferior belly of omohyoid
- c. Sternothyroid muscle
- d. Sternohyoid muscle
- e. Superior belly of omohyoid

**ANSWER: A**

Question 52

Excitation–contraction coupling in skeletal muscle involves all of the following events, EXCEPT

Select one:

- a. Increased conductance of sarcolemma to  $\text{Na}^{+}$
- b. Binding of  $\text{Ca}^{2+}$  to Calmodulin

- c. Conformational change in Dihydropyridine receptor
- d. ATP hydrolysis by myosin ATPase
- e. Depolarization of the T-tubule of the muscle membrane

**ANSWER: B**

#### Question 53

Which of the following is wrong regarding craniofacial defects?

Select one:

- a. Cranioschisis is seen due to a failure of sclerotomal cells to form the cranial vault.
- b. Craniofacial defects are common birth defects because neural crest cells are targets of teratogens.
- c. Cranial meningocele is a herniation of cranial meninges
- d. Cranial meningoencephalocele is less severe than anencephaly
- e. Exposure of the brain tissue to amniotic fluid can result anencephaly.

**ANSWER: A**

#### Question 54

One of the following is not the clinical use of H1 receptor antagonists

Select one:

- a. In atopic dermatitis
- b. In motion sickness
- c. In peptic ulcer disease
- d. In allergic rhinitis
- e. In drug induced parkinsonism

**ANSWER: C**

Question 55

False about muscular dystrophies

Select one:

- a. Deletion of dysrophin gene on both chromosomes is incompatible with life
- b. Carrier females have variable clinic course depending on x-chromosome inactivation
- c. Respiratory failure is common cause of death
- d. Inability to walk at age of one is classic in duchenne muscular dystrophy
- e. Deletion (complete absence) of dystrophin is responsible for baker's muscular dystrophy

**ANSWER: E**

**Question 56**

Which muscle attaches to the medial surface of the superior part of the tibia as part of the pes anserinus formation in conjunction with the tendinous insertions of the sartorius and gracilis?

Select one:

- a. semitendinosus
- b. adductor longus
- c. semimembranosus
- d. tensor fascia lata
- e. biceps femoris

**ANSWER: A**

Question 57

Which of the following is not a clinical manifestation of lepromatous leprosy?

Select one:

- a. Many bacilli are present in lesions
- b. There is no mutilation of the extremities, only disfiguration is common
- c. Patients are not reactive to lepromin
- d. Skin nodules are among its important characteristics
- e. Numerous deep lesions are concentrated in cooler areas of the body

**ANSWER: B**

Question 58

Wrong concerning aspirin

Select one:

- a. Low dose aspirin may decrease incident of myocardial infarction
- b. May exacerbate asthmatic attack
- c. May cause peptic ulcer disease; even GI bleeding
- d. Irreversible COX -1 selective inhibitor
- e. Low dose aspirin has antiplatelet activity

**ANSWER: D**

Question 59

Which of the following best describes pathway of tears from lachrymal glands to the nasal cavity?

Select one:

a. Lacrimal punctum . . . lacrimal lake . . . lacrimal papilla . . .

lacrimal sac . . . nasolacrimal duct

b. Lacrimal ducts . . . lacrimal lake . . . lacrimal canaliculi . . .

lacrimal sac . . . nasolacrimal duct

c. Lacrimal ducts ... lacrimal sac ... lacrimal canaliculi ...

lacrimal lake ... nasolacrimal duct

d. Lacrimal canaliculi . . . lacrimal lake . . . lacrimal ducts . . .

lacrimal sac . . . nasolacrimal duct

e. Lacrimal canaliculi . . . lacrimal sac . . . lacrimal ducts . . .

lacrimal lake . . . nasolacrimal duct

**ANSWER: B**

Question 60

All of the following muscles of the pharynx are innervated by pharyngeal plexus except:

Select one:

a. Middle and inferior constrictor muscle

b. Salpingopharyngeus

c. Superior constrictor muscle

d. Palatopharyngeus

**ANSWER: D**

Question 61

Lymphatic drainage from the tongue is by all the following nodes except one

Select one:

- a. Submental groups
- b. Inferior deep cervical groups
- c. Submandibular groups
- d. Deep parotid groups
- e. Superior deep cervical groups

**ANSWER: D**

#### Question 62

The surface markings of the sciatic nerve are from:

Select one:

- a. the ischial tuberosity to the adductor tubercle of the medial femoral condyle
- b. the ischial tuberosity to the apex of popliteal fossa
- c. the midpoint of a line between the ischial tuberosity and greater trochanter to the adductor tubercle of the medial femoral condyle
- d. the posterior superior iliac spine to the apex of popliteal fossa
- e. the midpoint of a line between the ischial tuberosity and greater trochanter to the apex of popliteal fossa

**ANSWER: E**

#### Question 63

Which of the following best describes pathway of parasympathetic fibers to the lachrymal gland?

Select one:

- a. CN V . . . ophthalmic branch of V1 ... lacrimal branch of V1

- b. CN VII . . . deep petrosal nerve . . . nerve of the pterygoid canal  
 . . . pterygopalatine ganglion . . . infratrochlear branch of V2 . . .  
 lacrimal branch of V1
- c. CN VII . . . greater petrosal nerve . . . otic ganglion . . .  
 infraorbital branch of V2 ... lacrimal branch of V1
- d. CN IX . . . lesser petrosal nerve . . . nerve of the pterygoid  
 canal . . . pterygopalatine ganglion . . . infraorbital branch of V2 .  
 . . lachrymal branch of V1
- e. CN VII . . . greater petrosal nerve . . . nerve of the pterygoid  
 canal . . . pterygopalatine ganglion . . . zygomatic branch of V2 . .  
 . lachrymal branch of V1

**ANSWER: E**

#### Question 64

In a normal healthy skeletal muscle, which of the following events directly occurs as a result of propagation of an action potential to the presynaptic membrane of  $\alpha$ -motor neuron?

Select one:

- a. Depolarization of the T tubule membrane
- b. Opening of voltage gated  $\text{Na}^+$  channels on the NMJ
- c. Contraction of the muscle
- d. Opening of voltage-gated  $\text{Ca}^{2+}$  channels in the presynaptic motor neuron.
- e. Increase in intracellular  $\text{Na}^+$  concentration in the motor neuron terminal

**ANSWER: D**

**Question 65**

Regarding the popliteal fossa, which of the following is INCORRECT?

Select one:

- a. tibial nerves run along the midline of the fossa and disappears in between the heads of gastrocnemius
- b. the common peroneal nerve disappears into the substance of peroneal longus to lie on the neck of the fibula
- c. the superolateral border is formed by biceps femoris muscle
- d. at all levels in the popliteal fossa, the popliteal artery lies between the vein and nerve
- e. the roof of the fossa formed by fascia lata and pierced by small saphenous vein and the posterior femoral cutaneous nerve

**ANSWER: D**

**Question 66**

Of the following molecules which one is associated with the ability of *Staphylococcus aureus* to evade phagocytosis?

Select one:

- a. protein A
- b. enterotoxin
- c. beta-lactamase
- d. DNase
- e. Hyaluronidase

**ANSWER: A**

Question 67

Pale pink amorphous masses surrounded by a foreign body type granulomatous reaction in routine H and E stained sections of tissues sampled around joints is likely under light microscope to

Select one:

- a. Osteoarthritis
- b. Rheumatoid arthritis
- c. Gouty arthritis
- d. Tuberculous arthritis

**ANSWER: C**

Question 68

Regarding the innervation of muscles in the sole of foot, which statement is FALSE?

Select one:

- a. adductor hallucis supplied by the lateral plantar nerve
- b. all interossei are innervated by lateral plantar nerve
- c. quadratus plantae is supplied by the medial plantar nerve
- d. first lumbrical is supplied by the medial plantar nerve
- e. flexor digitorum brevis is supplied by the medial plantar nerve

**ANSWER: C**

Question 69

One of the following drugs should not be used in management of glaucoma

Select one:

- a. Tropicamide
- b. Epinephrine
- c. Pilocarpine
- d. Timolol
- e. Latanoprost

**ANSWER: A**

#### Question 70

Which statement about slow twitch fibers is correct?

Select one:

- a. Scarcemitochondria, Extensive capillary supply, low concentrations of myoglobin and Fatigue resistant
- b. Abundant mitochondria, Extensive capillary supply, low concentrations of myoglobin and fatigue resistant
- c. Abundant mitochondria, Extensive capillary supply, High concentrations of myoglobin and Fatigue resistant
- d. Scarcemitochondria, Extensive capillary supply, High concentrations of myoglobin and Fatigue resistant
- e. Abundant mitochondria, Extensive capillary supply, High concentrations of myoglobin and easyfatigue

**ANSWER: C**

#### Question 71

Select a wrong statement about tongue development?

Select one:

- a. Mesoderm from second, third and fourth pharyngeal arches forms root of tongue
- b. Body of the tongue is formed by lingual swellings of first pharyngeal arch
- c. Terminal sulcus separates anterior and posterior parts of tongue.
- d. Myotomes of the occipital somites form tongue muscles
- e. Mucus of the posterior tongue is supplied by the lingual nerve.

**ANSWER: E**

#### Question 72

One of the following is not a correct description of immunological and pathological spectrum of cutaneous leishmaniasis:

Select one:

- a. Natural healing is a common condition in patients with muco-cutaneous leishmaniasis
- b. There is a notable immunological spectrum behind the clinical variants characterized by humoral and cellular immune responses,
- c. Both parasite and host factors play a role in outcomes of disease variations,
- d. Similar treatments may be used in all variants, but with varying regimen,
- e. Some species of leishmania are known to depict the spectrum more clearly,

**ANSWER: A**

#### Question 73

What is the rationale that onchocerciasis can be eliminated?

Select one:

- a. There are effective treatments against the adult and larval stages of the parasite,
- b. Repeated mass administrations of microfilaricidal treatments can stop transmission to black flies,
- c. Currently available treatments are capable of altering the feeding behavior of black flies,
- d. Repeated doses of adulticide treatments reduces fecundity, and stop transmission to black flies,
- e. There is no drug resistance to currently available filaricidal drugs

**ANSWER: \_\_\_\_**

Question 74

Beta 2 agonist of choice for nocturnal asthma

Select one:

- a. Terbutaline
- b. Albuterol
- c. Salmeterol
- d. Metaproterenol
- e. Metoprolol

**ANSWER: C**

Question 75

The wrong association about the sensory innervation of the gum (proper gingivae) is

Select one:

- a. Upper lingual gingivae ----- greater palatine nerve(V2)
- b. Lower buccal gingivae----- posterior superior alveolar nerve (V2)
- c. Lower buccal gingivae ----- buccal nerve (V3)
- d. Lower lingual gingivae-----lingual nerve(V3)
- e. Upper labial gingivae ----- anterior superior alveolar nerve (V2)

**ANSWER: B**

#### Question 76

One of the following is a wrong association relations of walls of the tympanic cavity

Select one:

- a. Medial wall ----- labyrinthine promontory
- b. Floor ----- internal carotid artery
- c. Posterior wall ----- pyramidal prominence
- d. Anterior wall ----- tensor tympani muscle
- e. Lateral wall ----- manubrium of malleus

**ANSWER: B**

#### Question 77

All of the following are associated with cutaneous diseases caused by *Staphylococcus aureus* EXCEPT \_\_\_\_\_.

Select one:

- a. Impetigo

- b. Folliculitis
- c. Abscess
- d. Gas gangrene
- e. Carbuncles

**ANSWER: D**

Question 78

Wrongly matched; serotonin receptor agonist: clinical use

Select one:

- a. Fluoxetine: in depression
- b. Ondansetron: chemotherapy induced prevention of nausea and vomiting
- c. Cisapride: gastroesophageal reflux disorders
- d. Buspirone: anxiolytic
- e. Triptans: in migraine headache

**ANSWER: B**

Question 79

Second generation antihistamine

Select one:

- a. Fexofenadine
- b. Chlorpheniramine
- c. Promethazine
- d. Diphenhydramine
- e. Hydroxyzine

**ANSWER: A**

Question 80

Which one of the following action of muscle of larynx is lost due to injury to external laryngeal nerve?

Select one:

- a. Relaxing the vocal cords
- b. Narrowing the laryngeal inlet
- c. Adducting the vocal cords
- d. Abducting the vocal cords
- e. Tensing the vocal cords

**ANSWER: E**

Question 81

With regard to the muscular system, which statement is false?

Select one:

- a. Individual skeletal muscle cells are covered by endomysium
- b. Smooth muscle cells may be found as isolated single cells
- c. Cardiac T tubules are found at the level of the A-I junction
- d. Skeletal muscle is composed of large, elongated, multinucleated cells
- e. Cardiac muscle cells branch and anastomose

**ANSWER: C**

Question 82

Which one is incorrect regarding external laryngeal nerve?

Select one:

- a. Supply superior thyroid artery
- b. Lies on the inferior constrictor muscle of the pharynx
- c. Terminal branch of the superior laryngeal nerve
- d. Is purely motor
- e. Supplies thyrohyoid muscle

**ANSWER: E**

Question 83

Which of the following is most likely associated with rigor mortis

Select one:

- a. Digestion of muscle proteins by lysosomes
- b. Depletion of ATP
- c. Increased myosin ATPase activity
- d. Damage to actin and myosin filaments
- e. Rapid sequestration of  $\text{Ca}^{2+}$  back to the SR

**ANSWER: B**

Question 84

Which of the following muscles is NOT innervated by the hypoglossal nerve?

Select one:

- a. Palatoglossus
- b. Intrinsic muscles of the tongue
- c. Styloglossus
- d. Genioglossus
- e. Hyoglossus

**ANSWER: A**

**Question 85**

Which one of the following is not feature of typical cervical vertebra?

Select one:

- a. The presence of odontoid process
- b. Bifid spinout process
- c. Large and triangular vertebral foramen
- d. Foramen transversarium in transverse process
- e. Relatively small and broad body

**ANSWER: A**

**Question 86**

Which one of the following statement is true?

Select one:

- a. Glycogen is the primary fuel during exercise lasting 60-90 minutes
- b. Blood free fatty acids are the primary fuel molecules during exercise lasting 60-90 minutes
- c. VLDL is the primary fuel during exercise lasting 60-90 minutes
- d. Blood glucose primary fuel during exercise lasting 60-90 minutes
- e. Amino acids are primary fuel molecules during exercise lasting 60-90 minutes

**ANSWER: A**

**Question 87**

Which one of the following muscles with their innervation is CORRECTLY paired?

Select one:

- a. adductor longus ⇒ femoral nerve
- b. obturator externus ⇒ sciatic nerve
- c. obturator internus ⇒ pudendal nerve
- d. gluteus minimus ⇒ inferior gluteal nerve
- e. gracilis ⇒ obturator nerve

**ANSWER: E**

Question 88

All of the following are true regarding soft tissue tumors, Except:

Select one:

- a. Classification of tumors is according to their differentiation lines
- b. All originate from primitive mesenchymal stem cells
- c. Intermediate grade tumors are aggressive and have metastatic potential
- d. Lipoma The most common benign soft tissue tumor of adulthood
- e. None

**ANSWER: C**

Question 89

Correct concerning botulinum toxin

Select one:

- a. Inhibits acetylcholine release into the synaptic space
- b. Inhibits acetylcholinesterase enzyme
- c. Inhibits choline uptake
- d. Inhibits choline acetyltransferase enzyme

e. Inhibits the transport of acetylcholine into vesicles

**ANSWER: A**

Question 90

Mechanism of action of atomoxetine, which is usually used in management of attention deficit disorders

Select one:

- a. Inhibits MAO (monoamine oxidase) enzyme
- b. Increases the release of catecholamines from storage vesicle
- c. Inhibits the synthesis of catecholamines
- d. Inhibits vesiculation of catecholamines
- e. Inhibits NET (norepinephrine transporter)

**ANSWER: E**

Question 91

Which of the following is not true regarding giant cell tumor of bone?

Select one:

- a. Typical location is in the metaphysis of long bones
- b. Its behavior can be characterized as locally aggressive, non metastasizing
- c. It most commonly affects children under five years of age
- d. Osteoclasts represent the neoplastic cells of giant cell tumor
- e. None of the above

**ANSWER: C**

Question 92

All the following structures are related with labyrinthine(medial)

wall of tympanic cavity except:

Select one:

- a. Tympanic plexus
- b. Facial nerve
- c. Chorda tympani nerve
- d. Lesser petrosal nerve
- e. Tympanic nerve

**ANSWER: C**

#### Question 93

Regarding to the bone:

Select one:

- a. Volkmann's canal run perpendicular to the longitudinal axis of the bone
- b. Compact bone has outer and inner circumferential laminae known as the Haversian system
- c. Cell bodies of the osteocytes are found lining the surfaces of developing bone
- d. Spongy bone contains more Haversian system than the compact bone
- e. Canaliculi contain the cell bodies of the osteoblasts

**ANSWER: A**

#### Question 94

Select a wrong statement regarding development of a permanent teeth?

Select one:

- a. Start to form around 3rd month of development but stay dormant
- b. Located at labial aspect of the milk teeth
- c. Growth helps to shed the milk teeth
- d. Root of deciduous teeth is resorbed by osteoclasts
- e. None of the above

**ANSWER: B**

Question 95

To test the normal function of inferior oblique muscle of one eye, what would you have a patient do?

Select one:

- a. Elevate and laterally rotate the eye
- b. Elevate and adduct the eye
- c. Depress and abduct the eye
- d. Depress and adduct the eye
- e. Elevate and medially rotate the eye

**ANSWER: A**

Question 96

28 years old professional cyclist has been training for an opportunity to go for a long race. His coach strongly suggests the intake of carbohydrates after work out to ensure muscle glycogen storage. The activity of muscle glycogen synthesis in resting muscle is increased by the action of which of the following?

Select one:

- a. Norepinephrine
- b. Cortisol
- c. Epinephrine
- d. Glucagon
- e. Insulin

**ANSWER: E**

Question 97

\_\_\_\_\_ can be often used in patients who are unable to exercise during the cardiac stress test

Select one:

- a. Isoproterenol
- b. Dobutamine
- c. Norepinephrine
- d. Midodrine
- e. Methoxamine

**ANSWER: B**

**Question 98**

Keratinocytes die in which layer of epidermis?

Select one:

- a. Stratum granulosum
- b. Stratum corneum
- c. Stratum spinosum
- d. Stratum basale

e. Stratum lucidum

**ANSWER: A**

**Question 99**

Which of the following statements is correct about the cells of cartilage?

Select one:

- a. Chondrocytes have eosinophilic cytoplasm
- b. Chondroblasts are found in the lacunae
- c. Chondrocytes located peripherally are round, while those located deeper are fusiform
- d. Chondrogenic cells can differentiate into osteogenic cells
- e. Chondroblasts have processes which communicate by gap junction with each other

**ANSWER: D**

**Question 100**

Which of the following tongue muscles is correctly paired with its action?

Select one:

- a. Vertical ... narrows and elongates tongue
- b. Palatoglossus . . . depresses and retracts tongue
- c. Superior and inferior longitudinal . . . curls tip and sides of tongue and shortens tongue
- d. Genioglossus and hyoglossus . . . elevate posterior part of tongue
- e. Transverse . . . fattens and broadens tongue

**ANSWER: C**

Question 101

Which one of the following viruses is the leading cause of congenital malformations?

Select one:

- a. Fifth disease
- b. Cocksackie infection
- c. Polio
- d. Sixth disease
- e. Rubella

**ANSWER: E**

Question 102

The femoral nerve arises from which of the following segment of the spinal cord?

Select one:

- a. L2 and L3
- b. L4, L5, S1, S2 and S3
- c. L5, S1, S2 and S3
- d. L1 and L2
- e. L2, L3 and L4

**ANSWER: E**

Question 103

The site at which the three parts the hip bones fused:

Select one:

- a. acetabulum
- b. iliac crest
- c. iliac fossa
- d. obturator membrane
- e. symphysis pubis

**ANSWER: A**

Question 104

A diffuse rash associated with some *Streptococcus pyogenes* infections is a symptom of \_\_\_\_\_.

Select one:

- a. toxic shock syndrome toxic shock syndrome
- b. Abscess
- c. scarlet fever
- d. pharyngitis
- e. Cellulitis

**ANSWER: C**

Question 105

Best beta blocker to be used in asthmatic hypertensive patients

Select one:

- a. Propranolol
- b. Pindolol
- c. Atenolol
- d. Nadolol

e. Celiprolol

**ANSWER: E**

Question 106

Which nerve is NOT correctly matched with its distribution?

Select one:

- a. Chorda tympani nerve . . . motor fibers to tensor tympani muscle
- b. Lingual nerve . . . sensation from floor of mouth and lingual gingivae
- c. Chorda tympani nerve . . . secretomotor fibers to submandibular and sublingual glands
- d. Lingual nerve . . . sensation from the anterior two-thirds of the tongue
- e. Chorda tympani nerve . . . taste fibers from anterior two-thirds of tongue

**ANSWER: A**

Question 107

Which of the following muscles is NOT properly matched with its main action?

Select one:

- a. Lateral rectus . . . . . abducts eyeball
- b. Superior oblique . . . adducts, elevates, and rotates eyeball laterally
- c. Inferior rectus . . . . . depresses, adducts, and rotates eyeball medially
- d. Superior rectus . . . . . elevates, adducts, and rotates eyeball medially
- e. Inferior oblique . . . . . abducts, elevates, and rotates eyeball laterally

Question 108

Which of the following statements is true about the contractile

response of skeletal muscle

Select one:

- a. Skeletal muscle cells are normally held at a muscle length that maximizes their ability to generate greater force
- b. Decreased concentration of  $\text{Ca}^{2+}$  in the ECF highly affects the contractile response of the skeletal muscle.
- c. During isometric contraction, the tension goes up and the muscle shortens
- d. Contractile response can vary by different hormones
- e. Excitation of the Dihydropyridine receptor results in the influx of  $\text{Ca}^{2+}$  from the ECF

**ANSWER: A**

Question 109

One of the following drug is not a leukotriene antagonist

Select one:

- a. Zileuton
- b. Zafirlukast
- c. Montelukast
- d. Pranlukast
- e. Cromolyn

**ANSWER: E**

**Question 110**

Which of the following nerves supplies muscles derived from the 4th pharyngeal arch?

Select one:

- a. Mandibular
- b. Recurrent laryngeal
- c. Glossopharyngeal
- d. Facial
- e. Superior laryngeal

**ANSWER: E**

Question 111

Candida species can cause all of the following mucocutaneous infections except:

Select one:

- a. Paronychia
- b. Angular cheilitis
- c. Mucormycoses
- d. candida intertrigo
- e. Thrush

**ANSWER: C**

**Question 112**

With regard to cartilage, which statement is false?

Select one:

- a. Fibrocartilage is used for endochondral bone ossification
- b. Fibrocartilage has thick bundles of collagen fibres in the matrix
- c. Fibrocartilage is present in regions subjected to high tension
- d. Hyaline cartilage is characterized by an amorphous glassy matrix

e. Elastic cartilage is similar to hyaline cartilage except for the abundant elastic fibres it contains

**ANSWER: A**

Question 113

Epinephrine reversal is when epinephrine is given with

Select one:

- a. Alpha 1-AR antagonists
- b. Non selective alpha and beta agonists
- c. Alpha 1-AR agonists
- d. Beta 1-AR agonists
- e. Beta 1-AR antagonists

**ANSWER: A**

Question 114

Mr.KK, a known epileptic patient who has been taking phenytoin, was diagnosed with hypertension and was put on atenolol. What is the possible drug-drug interaction

Select one:

- a. Phenytoin increases the metabolism of atenolol
- b. Phenytoin displaces atenolol from plasma protein binding
- c. Phenytoin decreases the absorption of atenolol
- d. Phenytoin decreases the excretion of atenolol
- e. Phenytoin competes with atenolol at receptor site

**ANSWER: A**

**Question 115**

Which bony markings of femur found distally?

Select one:

- a. lesser trochanter
- b. greater trochanter
- c. adductor tubercle
- d. pectineal line
- e. gluteal tuberosity

**ANSWER: C**

Question 116

The presence of antibodies against cyclic citrullinated peptides in the serum of patients with joint disease is a diagnostic marker of

Select one:

- a. Gouty arthritis
- b. Ankylosing spondylitis
- c. Rheumatoid arthritis
- d. Osteoarthritis

**ANSWER:C**

Question 117

Possible adverse effect from high dose oxymetazoline

Select one:

- a. Hypertension

- b. Miosis
- c. Hypotension
- d. Increased intraocular pressure
- e. Sedation

**ANSWER: C**

**Question 118**

All of the following can be causes of cleft palate, EXCEPT:

Select one:

- a. Smallness of palatine shelves
- b. Lack of fusion of maxillary and medial nasal prominences
- c. Molecular inhibition of the fusion process itself
- d. Failure of the palatine shelves to elevate,
- e. Micrognathia blocking tongue to drop

**ANSWER: B**

**Question 119**

Black piedra is a fungus infection of the \_\_\_\_\_

Select one:

- a. Skin
- b. Nails
- c. Hair
- d. Foot

**ANSWER: C**

Question 120

Beta blocker with the shortest duration of action

Select one:

- a. Esmolol
- b. Acebutolol
- c. Bisoprolol
- d. Nebivolol
- e. Atenolol

**ANSWER: A**

**GOOD LUCK!**